

Eigenvalues and inverse problems

Johnson's derivative-realizability conjecture

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Challenging reflects preserving nonnegative realizability under polynomial differentiation; community impact connects the nonnegative inverse eigenvalue problem with polynomial critical points.

Problem statement

For every integer $n \geq 5$ and every real entrywise-nonnegative matrix $A \in \mathbb{R}^{n \times n}$, define $p_A(z) = \det(zI - A)$. Must there exist an entrywise-nonnegative $B \in \mathbb{R}^{(n-1) \times (n-1)}$ such that

$$\det(zI - B) = \frac{1}{n} p'_A(z) \quad \text{as polynomials in } z?$$

Equivalently, the critical points of the characteristic polynomial, counted with multiplicity, would themselves form a realizable spectrum of the smaller order. Neither A nor B is assumed symmetric or diagonalizable. The realization must have exactly order $n - 1$; allowing arbitrary additional zero eigenvalues changes the problem. This would provide a dimension-reduction operation for nonnegative spectral realization.

References

Hoover, McCormick, Paparella, and Thrall, *On the realizability of the critical points of a realizable list*, Conjecture 1.2, p. 2, and §6. The paper credits the conjecture to Johnson and records the Cronin–Laffey low-order results.

Earlier status check — 2026-09-08

Searches for *Johnson conjecture derivative nonnegative matrix characteristic polynomial proof counterexample*, 1712.05454 2026, and *Monov conjecture solved* found no general resolution. The source proves several classes and records the solved cases $n \leq 4$, and $n \leq 6$ with $\text{tr } A = 0$. Nonnegative power sums alone are a different hypothesis. Monov's weaker moment conjecture is not separately counted here.

Audit update — 2026-09-10

Rechecked the [primary manuscript](#), Conjecture 1.2 and its proved families; the [journal version](#) is LAA 555 (2018), 301–313. Results for Ciarlet/Suleĭmanova lists, appropriate companion-matrix realizations, and trace-zero lists of orders five and six settle substantive portions of the displayed target. Johnson-conjecture/critical-point searches found no general proof or counterexample.